Causal Inference Notation Guide

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STAT 579: Comprehensive Notation Guide (Final Version)

1. Custom LaTeX Macros

A well-defined preamble is key for efficiency. These macros provide shortcuts for commonly used mathematical notation.

```
% --- Core Symbols & Operators ---
\newcommand{\PP}{\mathbb{P}} % probability
\newcommand{\E}[1]{\mathbb{E}\left[#1\right]} % Expectation
\newcommand{\Es}[2]{\mathbb{E}_{#1}\left[#2\right]} % Expectation with subscript
\newcommand{\Prob}[1]{\mathbb{P}\left(#1\right)} % Probability with parentheses
\newcommand{\var}[1]{\operatorname{Var}\left[#1\right]}
\newcommand{\cov}[2]{\operatorname{Cov}\left(#1, #2\right)}
\newcommand{\indep}{\perp\!\!\!\perp} % Independence symbol
\newcommand{\doop}[1]{\operatorname{do}\left(#1\right)} % do-operator
\newcommand{\ind}[1]{\mathbb{I}\left\{#1\right\}}   % Indicator function

% --- Mathematical Sets & Vectors ---
\newcommand{\RR}{\mathbb{R}} % Real numbers
\newcommand{\NN}{\mathbb{N}} % Natural numbers
\newcommand{\vect}[1]{\mathbf{#1}} % Vector notation
```

2. Variables, Notation, and Their Support

At the heart of any causal analysis are key variables. It's crucial to distinguish between a **random variable** and a **fixed value**. The **support** of a random variable is the set of all possible values it can take.

Symbol	Type	Support ($\mathcal{A}, \mathcal{Y}$, etc.)	Description & Interpretation
i	**Fixed** Index	$1, \dots, n$	A single unit or individual in a sample of size n
A, A_i	**Random** Variable	$\mathcal{A}$, typically $0, 1$	**Treatment/Exposure**. The value of A_i is not known until it is observed
a	**Fixed** Value	$a \in \mathcal{A}$	A specific, hypothetical level of treatment (e.g., $a = 1$)
Y, Y_i	**Random** Variable	$\mathcal{Y} \subseteq \mathbb{R}$	**Outcome**. The variable we want to affect
y	**Fixed** Value	$y \in \mathcal{Y}$	A specific, hypothetical outcome value
X, X_i	**Random Variable**	$\mathcal{X} \subseteq \mathbb{R}$	**Covariates**. A variable of baseline characteristics (e.g., age) that may affect treatment assignment or the outcome
$\mathbf{X}, \mathbf{X}_i$	**Random Vector**	$\mathcal{X} \subseteq \mathbb{R}^d$	A **vector of baseline covariates**. Bold **X** signifies a collection of multiple variables (e.g., age, sex, baseline health)
Z, Z_i	**Random** Variable	$\mathcal{Z}$, typically $0, 1$	An **Instrumental Variable (IV)**
M, M_i	**Random** Variable	$\mathcal{M} \subseteq \mathbb{R}$	A **Mediator**. A post-treatment variable on the causal pathway
C, C_i	Explicit confounder variable	$\mathcal{C} \subseteq \mathbb{R}$	Variable explicitly used to distinguish from general covariates
$\mathbf{C}$	Vector of con-founders	$\mathcal{C} \subseteq \mathbb{R}^d$	Vector of explicit confounding variables
L, L_i	Time-indexed confounder	$\mathcal{L} \subseteq \mathbb{R}$	Time-indexed or longitudinal confounder (in MSM contexts)
$\mathbf{L}$	Vector of con-founders	$\mathcal{L} \subseteq \mathbb{R}^d$	Vector of time-indexed or longitudinal confounders (in MSM contexts)

3. The Potential Outcomes Framework

This framework forces us to think about counterfactual worlds to define causal effects.

Notation	How to Read It	Random vs. Fixed	Interpretation
Y_i^a	"The potential outcome `Y` for individual `i` under treatment `a`."	Y_i^a is a **random variable**. The superscript `a` is a **fixed value**.	This represents the outcome that *would have been observed* for person `i` if their treatment had been set to level `a`.
A_i^z	"The potential treatment `A` for individual `i` under instrument `z`."	A_i^z is a **random variable**. The superscript `z` is a **fixed value**.	This is the treatment person `i` *would have* if their instrument `Z` were set to `z`.
$Y^{a,M^{a'}}$	"Y under treatment `a` and the mediator value that *would have occurred* under treatment `a'`."	$Y^{a,M^{a'}}$ is a **random variable**. Superscripts `a` and `a'` are **fixed**. $M^{a'}$ is a **random variable**.	A nested counterfactual for mediation. It asks: what would `Y` be if we set `A` to `a` but forced `M` to the value it would have naturally taken if `A` had been `a'`?

4. Fundamental Causal Assumptions

No causal conclusion is possible without untestable assumptions. Being explicit about them is paramount.

Assumption	Notation / Name	Explanation
Consistency	If $A_i = a$, then $Y_i = Y_i^a$.	The observed outcome for an individual is their potential outcome corresponding to the treatment they actually received. This links the observed data to the counterfactual world.
SUTVA	Stable Unit Treatment Value Assumption	Two parts: (1) **No interference**: My treatment doesn't affect your outcome. (2) **One version of treatment**: The effect of the treatment is the same for everyone who receives it.
Exchangeability	$Y^a \perp\!\!\!\perp A$	The potential outcomes are independent of the treatment that was *actually assigned*. This is the "as if random" assumption that holds in an ideal RCT.
Conditional Exchange-ability	$Y^a \perp\!\!\!\perp A \mid \mathbf{X}$	Within levels of the covariates $\mathbf{X}$, treatment assignment is random. This is the key assumption for observational studies, also called **ignorability** or **no unmeasured confounding**.

Assumption	Notation / Name	Explanation
Positivity	$0 < \mathbb{P}(A = a \mid \mathbf{X} = \mathbf{x}) < 1$	For every group of individuals with the same covariate values $\mathbf{x}$, some received treatment and some did not. There are no groups with a 0% or 100% chance of treatment.

5. Directed Acyclic Graphs (DAGs)

DAGs are visual representations of our causal assumptions, providing a powerful way to assess confounding. A DAG consists of nodes (variables) and directed edges (arrows) representing causal effects.

Concept	Description	Example
Path	A sequence of connected nodes. Can go along or against arrows.	$A \to M \to Y$ is a causal path. $A \leftarrow X \to Y$ is a non-causal path.
Confounder	A common cause of A and Y.	A variable X with paths $X \to A$ and $X \to Y$. This creates a "backdoor" path.
Mediator	A variable M on the causal pathway from A to Y.	A variable M in the path $A \to M \to Y$.
Collider	A common *effect* of two other variables.	A variable C in a path like $A \to C \leftarrow Y$. Conditioning on a collider *opens* a non-causal path and induces bias.
d-separation	A graphical rule for determining if variables are independent. Two variables are d-separated if all paths between them are blocked. A path is blocked if it contains a non-collider that is conditioned on, or a collider that is *not* conditioned on.	

Identification With DAGs

- **The do-operator:** $\mathbb{E}[Y \mid \text{do}(A = a)]$ represents the expectation of Y in a hypothetical world where we **intervene** and set `A=a` for everyone. This is mathematically equivalent to the potential outcome expectation $\mathbb{E}[Y^a]$.
- **Backdoor Criterion:** The causal effect of A on Y is identified if we can find a set of covariates $\mathbf{X}$ that blocks all non-causal ("backdoor") paths from A to Y. If $\mathbf{X}$ satisfies this, then:

$$\mathbb{E}[Y^a] = \sum_{\mathbf{x}} \mathbb{E}[Y \mid A = a, \mathbf{X} = \mathbf{x}] \, \mathbb{P}(\mathbf{X} = \mathbf{x})$$

6. Causal Mediation Analysis

Mediation analysis decomposes the total effect of A on Y into a **direct effect** and an **indirect effect** that travels through a mediator M.

Estimands

- **Natural Direct Effect (NDE):** The effect of changing treatment from 0 to 1 while forcing the mediator to stay at the level it would have naturally taken under no treatment.

$$\text{NDE} = \mathbb{E}\left[Y^{a=1,M^{a=0}} - Y^{a=0,M^{a=0}}\right]$$

- **Natural Indirect Effect (NIE):** The effect of holding treatment fixed at 1, but changing the mediator from its natural level under control to its natural level under treatment.

$$\text{NIE} = \mathbb{E}\left[Y^{a=1,M^{a=1}} - Y^{a=1,M^{a=0}}\right]$$

- **Total Effect (TE):** $\text{TE} = \text{NDE} + \text{NIE} = \mathbb{E}\left[Y^1 - Y^0\right]$

Mediation Assumptions (The "4-Way" Assumptions)

To identify the NDE and NIE, we need four conditional exchangeability assumptions (no unmeasured confounding) for the following relationships, conditional on baseline covariates $\mathbf{X}$:

1. The effect of treatment A on the outcome Y.
2. The effect of treatment A on the mediator M.
3. The effect of the mediator M on the outcome Y.
4. There is no confounder of the $M - Y$ relationship that is itself affected by the treatment A.

7. Estimation Strategies

7.1 Standardization (G-Formula)

- **Goal:** Estimate $\mathbb{E}[Y^a]$ by modeling the outcome.
- **Formula:**

$$\mathbb{E}[Y^a] = \sum_x \mathbb{E}[Y \mid A = a, \mathbf{X} = \mathbf{x}]\, \mathbb{P}(\mathbf{X} = \mathbf{x}) = \mathbb{E}_{\mathbf{X}}[\mathbb{E}[Y \mid A = a, \mathbf{X} = \mathbf{x}]]$$

7.2 Inverse Probability Weighting (IPW)

- **Goal:** Create a "pseudo-population" where treatment assignment is independent of $\mathbf{X}$.
- **Propensity Score:** The probability of receiving treatment: $e(\mathbf{X}) = \mathbb{P}(A = 1 \mid \mathbf{X})$.
- **IP Weight (for ATE):** The weight for each individual i is the inverse of their probability of receiving the treatment they actually got:

$$w_i = \frac{A_i}{e(\mathbf{X}_i)} + \frac{1 - A_i}{1 - e(\mathbf{X}_i)}$$

- **ATE Estimator:**

$$\mathbb{E}[\hat{Y}^a] = \frac{1}{n} \sum_{i=1}^{n} \frac{\mathbb{I}\{A_i = a\}\, Y_i}{\mathbb{P}(A_i = a \mid \mathbf{X}_i)}$$

More on Nested Counterfactual in Mediation Analysis

While the parenthesis notation $Y(a, M(a'))$ is arguably more explicit, the superscript notation $Y^{a,M^{a'}}$ is also widely used, particularly in the tradition of Pearl's work. The key is to be very precise in our interpretation.

Here is the revised section using the superscript notation, with a detailed breakdown to ensure clarity.

Deconstructing Mediation: A Guide to Superscript Notation

The nested potential outcome notation used in mediation, $Y^{a,M^{a'}}$, is powerful and compact, but it requires careful interpretation to distinguish between direct interventions and natural counterfactual values.

Let's break it down to fully understand its meaning. Think of the superscript as having two distinct components that describe a complex hypothetical world.

- **The First Component: A Direct Intervention on A**
  - The a in the superscript represents a direct **intervention**. We are imagining a world where we actively set the treatment A to the fixed value a.
- **The Second Component: A Counterfactual Value for M**
  - The $M^{a'}$ part is more subtle. This does **not** mean we are intervening on M. Instead, $M^{a'}$ represents the **potential value of the mediator**.
  - **How to Read $M^{a'}$**: "The value that the mediator M would have naturally taken if treatment A had been set to a'." This is a random variable, representing a natural outcome under a hypothetical condition.

Putting It All Together: $Y^{a,M^{a'}}$

The full expression combines these two ideas:

> "The potential outcome Y that would be realized if we **intervened** to set treatment A to a, while the mediator M simultaneously took on the value it would have **naturally** had if treatment had been set to a'."

This notation describes a complex counterfactual scenario where one variable (A) is set by intervention, while another variable (M) is determined by a separate counterfactual condition.

Revisiting The Causal Estimands with Superscript Notation

This detailed interpretation makes the definitions of the direct and indirect effects clear.

- **Natural Direct Effect (NDE)**
  - **Formula**: $\text{NDE} = \mathbb{E}\left[Y^{a=1,M^{a=0}} - Y^{a=0,M^{a=0}}\right]$
  - **Interpretation**: We compare two hypothetical worlds. In both, the mediator is held to the value it would have *naturally taken under the control condition* ($M^{a=0}$). The only difference is our direct intervention on A, changing it from 0 to 1. Any resulting change in the average outcome is therefore the direct effect of A that does not pass through M.
- **Natural Indirect Effect (NIE)**
  - **Formula**: $\text{NIE} = \mathbb{E}\left[Y^{a=1,M^{a=1}} - Y^{a=1,M^{a=0}}\right]$
  - **Interpretation**: Here, our direct intervention on A is held constant at the treated level ($a = 1$) in both scenarios. The only thing we change is the mediator's value—from the natural level it would have under control ($M^{a=0}$) to the natural level it would have under treatment ($M^{a=1}$). Any change in the average outcome must be due to the causal pathway that is transmitted *through* the mediator.